HIGH-LEVEL DESIGN

System Architecture

ATLANTECH · INTELLIGENT PORT OPERATIONS

Modular Monolith + Python Satellites

One dedicated on-premises deployment per port · same stack for every port

v3.0 · aligned with the UML set (UC-1–3 · 1.1–3.4 · AD-1–20 · SM-1–6)

PORT SERVER · ON PREMISES IN THE PORT LAN · ONE PER PORT

Office users · authority · departments · terminals

Web Dashboard
Vue.js · inside the port LAN

HTTPS · SSE

Field crews · pilots · tug masters · supervisors

Mobile App
Flutter · offline-first Outbox

HTTPS · WebSocket

EXTERNAL
MDM Platform
managed devices · enroll · wipe

NGINX
load balancer · TLS 1.3 termination

VPN Gateway
per-app always-on · split DNS app.atlantech.io

REST · SSE · /sync/events

LARAVEL MODULAR MONOLITH · PORT API

App Instance 1 | **App Instance 2**

every write: auth 401 → role & terminal 403 → app version 426 → license mode 423 → version 409 → rules 409/422 → plan limits 422 → audit

FUNCTIONAL AREAS

- Auth & RBAC
- License & Mode
- Check-in Client
- Users & Devices
- Port & Resources
- Berthing Requests
- BAPLIE
- Recommendations
- Plans & Dispatch
- Container Yard
- Mobile Sync
- Tracking & Deviations
- Notifications
- Audit

Realtime Server
Reverb · SSE → web · WebSocket → mobile
sent only after the DB commit

Scheduler
weather & sea level · every 6 h
astronomical tide · daily
license check-in · daily
Part 3 evaluation & retraining
task re-send T+10/T+20 · escalate T+30

cache · jobs

upload

periodic

REDIS cache · hot state · one job queue per worker

- queue:part1
- queue:part2
- queue:edi

BAPLIE File Volume
dedicated local storage

read file

writes → primary · reads ← replica

PYTHON SATELLITES · WORKERS

AI PART 1
Part 1 · Predictive & Env.
weather, wave & sea level
primary + secondary provider
astronomical tide (local, daily)
→ feasible entry-time windows
constraints only · never allocates

AI PART 2
Part 2 · AI Engine
service-time & delay-risk prediction
CP-SAT · berth · tug · pilot
FIFO fallback on solver failure
yard placement (CT)
template rationale · not an LLM

WORKER
EDI Parser
BAPLIE D.17A
deterministic · not AI
no partial commit on errors
parse_status PARSED / QUARANTINED

AI PART 3
Part 3 · Feedback Loop
trusted data only · drift watch
forecast bias correction
guarded retraining
deploy only if not worse
rollback model kept · local only

read port state · write results

MYSQL · PORT OPERATIONAL DATABASE

Primary
all writes

replication

Replica
dashboard reads

terminal-level row security
append-only audit log
AES-256 at rest

OUTSIDE THE PORT SERVER

EXTERNAL
Weather & Marine Providers
primary + secondary · every 6 h + on demand

MANUAL
Terminal Operating System (TOS)
not integrated · CT Operation Planner exports the yard plan and enters it into the TOS by hand

Not connected by design
no external request source (staff enter requests)
no payment gateway · no push service

LICENSE ONLY

NO OPERATIONAL DATA · NO AI MODELS CROSS

CENTRAL SaaS · PLATFORM & LICENSING

SaaS Admin · password + email one-time code

SaaS Dashboard
Vue.js

SaaS API
Laravel · admins · plans · governorates
ports · licenses · billing records

daily check-in · outbound
HTTPS · signed token back

Check-in API
daily port check-in
returns signed token

License Signer
KMS-held key · kid
signs tokens lazily

Registry DB · MySQL

- admin accounts
- governorates
- license plans
- port registry & lifecycle
- annual licenses
- signing keys
- billing records
- platform audit

never stores operational or training data

ATLANTECH Support Engineer
deploys the signed build · bootstraps the port
signs the port TLS certificate · offline Root CA
support session opened & closed by Port IT

Manual Token File
for ports without internet · ACTIVE license only
delivered out of band · uploaded by Port Admin

Port-side license mode
FULL · port ACTIVE + license ACTIVE + valid token
READ_ONLY · any other case, always with a reason
SETUP · port ONBOARDING (setup writes only)
token reused while > 14 days remain;
banner warns at ≤ 14 days

LEGEND — request / response · critical write path · async job · scheduled run · license · external / manual / support link · AI satellites · databases · queues · licensing · scheduling

Smarter coordination. Stronger ports.